Math 133: Quiz 2

Friday, September 18, 2026 25 minutes 10 points

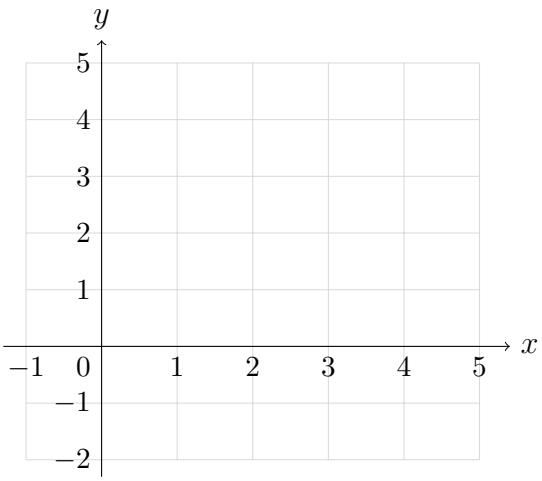
Name: _____

Show your work and include units when appropriate. Use algebra and graphs; derivatives are not needed. A simple scientific calculator is allowed.

1. (2 points) Let $f(x) = x^2 - 4x + 3$.

   (a) Find the vertex and both x-intercepts.

   (b) Sketch f on the grid. Label the vertex and all intercepts.



2. (2 points) Use $f(x) = x^2 - 4x + 3$ from Problem 1. For each interval, find the minimum and maximum values of f and give all x-values where they occur. Show the values you compare.

   (a) $0 \le x \le 4$.

   (b) $3 \le x \le 4$.

3. (2 points) A shop models its daily profit by

$$P(q) = 40 - (q - 6)^2,$$

where q is the number of gift baskets sold and $P(q)$ is in dollars.

(a) The shop can sell at most 4 baskets per day. State the allowed values of q. How many baskets should it sell to maximize profit, and what is that profit? Explain briefly.

(b) If the capacity increases to 8 baskets per day, how many should it sell to maximize profit, and what is that profit?

4. (2 points) A machine's resale value is modeled by

$$V(t) = 2000(0.90)^t, \qquad 0 \le t \le 5,$$

where t is years after purchase and $V(t)$ is in dollars.

(a) What do the numbers 2000 and 0.90 represent? What percentage of its value does the machine lose each year?

(b) Find $V(2)$ and interpret your answer in a sentence.

5. (2 points) Use the resale-value model from Problem 4.

(a) Find the dollar loss during the first year and the dollar loss during the second year.

(b) A buyer says, "The machine loses the same number of dollars each year." Is this correct? Explain using your answers and the model.